

# OCTO-SMART, INK-REDIBLE

OCTOPUS, THE MOST INTELLIGENT INVERTEBRATE

## SKIN CELLS

Chromatophores expand or contract to show colors like brown, red, and yellow  
Iridophores reflect light to create shiny or iridescent effects  
Leucophores reflect ambient light to match the surroundings



## MAZE-FINDING

Arms work independently, and undergo tactical exploration, allowing it to navigate, learn and remember its surroundings

## STRATEGIC SHELTER

Octopus use any material in the sea: coconut shells, cans, battery packs to shield their soft bodies from predators on the ocean floor where shelter is scarce



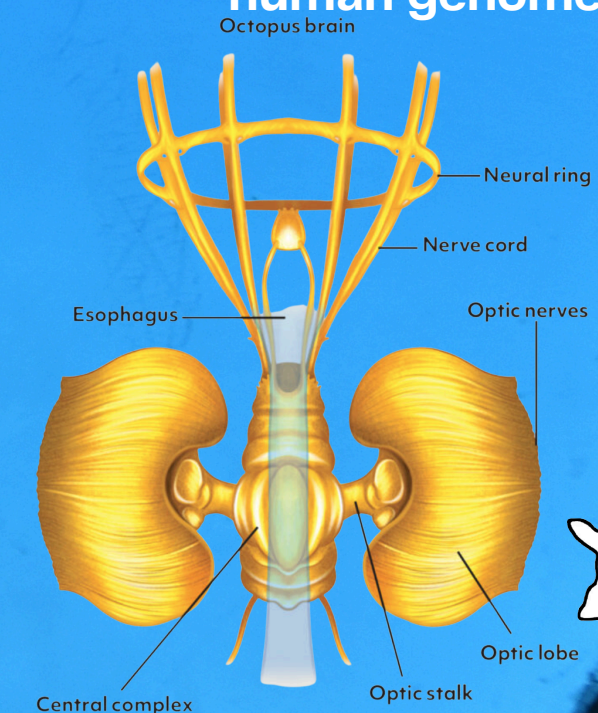
## ANIMAL IMITATION

The boneless body of the octopus allows it to change its shape. It's the first invertebrate species known that can systematically copy multiple species' appearances.

## THE BRAIN

Works using chemical signals, where editing microRNAs and RNA fine-tunes gene activity without changing genes.

The octopus genome shares genetic features with the human genome, including "Jumping genes", which account for approximately 45% of the human genome.



## NEURONS

Contains 500 million neurons, tasting, touching and controlling basic movements independently

